Teleology as Covariant Boundary Selection in a Consciousness–Ethics Field Theory

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Abstract

We present a mathematically disciplined implementation of teleology in a relativistic quantum field theory augmented by two scalar fields: a consciousness field $\Phi_c(x)$ and an ethical field $E(x)$. Instead of introducing an explicitly time-asymmetric bulk term in the Lagrangian, we encode teleological bias through a covariant boundary functional (or, equivalently, a covariant final weighting operator) that selects among globally unitary histories. Quantum evolution is formulated on arbitrary spacelike hypersurfaces via Tomonaga–Schwinger dynamics, thereby preserving local Lorentz covariance and microcausality. Measurement outcome statistics arise from a teleology-weighted decoherence functional, yielding an ethically biased Born rule as a special case while maintaining unitary evolution in the bulk. We state sufficient operator-algebraic conditions under which operational no-signaling is preserved despite teleology-induced reweighting.

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1 Motivation and Scope

Standard quantum field theory (QFT) and general relativity (GR) are locally Lorentz-covariant and (in the absence of measurement updates) fundamentally unitary. A teleological modification—a bias toward histories that increase some notion of ethical coherence or conscious order—is often implemented by adding an explicitly time-asymmetric term to the bulk action. Such terms are easy to write but hard to defend structurally: they appear as local violations of time-reversal symmetry that risk conflicting with the relativistic architecture of field theory.

This paper develops an alternative that is both conservative (bulk dynamics stays standard and unitary) and structurally explicit: teleology is represented by a covariant *boundary selection functional* at a future hypersurface, or equivalently by a covariant *final weighting operator* used only in the rule that maps amplitudes to outcome probabilities. The resulting framework can reproduce an "ethically biased Born rule" of the form

$$P(i) \propto |c_i|^2 \exp\big(-\mathcal{E}_i/C\big), \tag{1}$$

while maintaining unitary Tomonaga–Schwinger evolution for states on arbitrary spacelike hypersurfaces.

2 Classical Covariant Field Theory with Consciousness and Ethics

2.1 Fields and bulk action

Let $(M, g_{\mu\nu})$ be a globally hyperbolic spacetime. We consider the usual GR + Standard Model sector plus two additional scalar fields:

- $\Phi_c(x)$: a consciousness field (taken to be a scalar or multiplet; see Remark 1).
- $E(x)$: an ethical field encoding a local ethical potential or value density.

A minimal covariant bulk action is

$$S_{\text{bulk}} = \int_M \mathrm{d}^4x\,\sqrt{-g}\,\Big(\mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SM}} + \mathcal{L}_{\Phi_c} + \mathcal{L}_E + \mathcal{L}_{\text{int}}\Big), \tag{2}$$

where

$$\mathcal{L}_{\text{GR}} = \frac{1}{16\pi G}(R - 2\Lambda), \tag{3}$$

$$\mathcal{L}_{\Phi_c} = \frac{1}{2} g^{\mu\nu}\nabla_\mu\Phi_c\,\nabla_\nu\Phi_c - V_c(\Phi_c), \tag{4}$$

$$\mathcal{L}_E = \frac{1}{2} g^{\mu\nu}\nabla_\mu E\,\nabla_\nu E - V_E(E), \tag{5}$$

with V_c, V_E chosen (for example) as renormalizable quartic potentials.

2.2 Teleology as a boundary functional (structural implementation)

Instead of adding a time-asymmetric *bulk* term, we encode teleology via a *future boundary functional* at a spacelike hypersurface Σ_f:

$$S_{\text{tot}} = S_{\text{bulk}} + \lambda\,\Gamma[\Phi_c, E]\big|_{\Sigma_f}, \qquad \Gamma[\Phi_c, E]\big|_{\Sigma_f} := \int_{\Sigma_f} \mathrm{d}^3y\,\sqrt{h}\,\mathcal{T}(\Phi_c(y), E(y)). \tag{6}$$

Here h is the determinant of the induced metric on Σ_f and $\mathcal{T}$ is a scalar "teleology density."

The bulk Euler–Lagrange equations remain unchanged; teleology enters through modified boundary conditions. For a generic field $\varphi \in \{\Phi_c, E\}$ with canonical momentum density $\pi^\mu_\varphi := \partial\mathcal{L}/\partial(\nabla_\mu\varphi)$, variation yields

$$(\pi^\mu_\varphi n_\mu)\big|_{\Sigma_f} = \lambda\,\frac{\delta\mathcal{T}}{\delta\varphi}\Big|_{\Sigma_f}, \tag{7}$$

where n_μ is the future-directed unit normal on Σ_f. Equation (7) makes teleology structural: the local field equations are conventional; the *space of global solutions* is biased by boundary selection.

2.3 A bounded teleology density

To avoid runaway pathologies (e.g., "maximize E to $+\infty$"), it is useful to choose $\mathcal{T}$ to be bounded and to couple ethics to a scalar measure of consciousness organization. One example is

$$\mathcal{T}(\Phi_c, E) := \tanh\Big(\frac{E}{E_0}\Big)\,\mathcal{Q}(\Phi_c), \tag{8}$$

where $E_0 > 0$ sets a saturation scale and $\mathcal{Q}(\Phi_c) \geq 0$ is a Lorentz scalar "qualia density" functional. A minimal placeholder is $\mathcal{Q}(\Phi_c) = \Phi_c^2$; more structured choices are possible.

Remark 1 (Topology and "topological qualia)**.** *If phenomenology demands stable topological invariants ("topological qualia"), a single real scalar in $3+1$ dimensions is often insufficient to support nontrivial conserved topological charges. A technically natural upgrade is to take Φ_c as a complex scalar or an $O(N)$ multiplet, allowing conserved currents and/or topological densities. In that case, one can define a covariant current $j^\mu_{\Phi_c}$ and set $\mathcal{Q} = \sqrt{j_\mu j^\mu}$ or use hypersurface flux $\int_{\Sigma_f} \mathrm{d}\Sigma_\mu\, j^\mu$ inside Γ.*

3 Quantum Theory on Hypersurfaces: Unitary, Covariant Dynamics

3.1 Tomonaga–Schwinger evolution

Let Σ be an arbitrary spacelike Cauchy hypersurface and $|\Psi(\Sigma)\rangle$ the quantum state defined on Σ. The hypersurface-dependent (interaction-picture) evolution is given by the Tomonaga–Schwinger equation

$$\frac{\delta}{\delta\Sigma(x)}\,|\Psi(\Sigma)\rangle = -\mathrm{i}\,\hat{\mathcal{H}}_{\text{int}}(x)\;|\Psi(\Sigma)\rangle\,, \tag{9}$$

where $\hat{\mathcal{H}}_{\text{int}}(x)$ is a local Hamiltonian density operator.

A standard sufficient condition for foliation-independence of the evolution operator is microcausality:

$$[\hat{\mathcal{H}}_{\text{int}}(x), \hat{\mathcal{H}}_{\text{int}}(y)] = 0 \quad \text{for spacelike-separated } x, y. \tag{10}$$

Under (10), the unitary map between hypersurfaces depends only on the spacetime region they bound, not on the particular foliation used.

3.2 Teleology as a final weighting operator

We now implement teleology not by altering (9) (which would break unitarity) but by modifying the *rule that assigns probabilities* to histories/outcomes.

Let ρ_i be the initial density operator on Σ_i and define a final weighting operator on Σ_f:

$$\rho_f := \frac{1}{Z_f} \exp\left(\lambda\, \hat{\Gamma}\right), \qquad Z_f := \mathrm{Tr} \exp\left(\lambda\, \hat{\Gamma}\right). \tag{11}$$

Here $\hat{\Gamma}$ is the operator version of the boundary functional in (6), e.g.

$$\hat{\Gamma} = \int_{\Sigma_f} \mathrm{d}^3 y\, \sqrt{h}\, \hat{\mathcal{T}}(\hat{\Phi}_c(y), \hat{E}(y)). \tag{12}$$

Importantly, all evolution between Σ_i and Σ_f remains unitary. Teleology enters only through ρ_f.

4 Teleology-Weighted Histories and Outcome Probabilities

4.1 Class operators and decoherence functional

Consider a set of coarse-grained histories labeled by α. Let C_α denote the corresponding class (chain) operator, constructed from projectors/POVM elements localized in spacetime regions R_k and unitary evolution between them (formally definable using Tomonaga–Schwinger evolution). For decoherent histories, a probability assignment is obtained from a decoherence functional.

Define the *teleology-weighted decoherence functional*

$$D_T(\alpha, \beta) := \mathrm{Tr}\left(\rho_f\, C_\alpha\, \rho_i\, C_\beta^\dagger\right). \tag{13}$$

If off-diagonal interference terms are negligible in an appropriate coarse-graining (i.e., $D_T(\alpha, \beta) \approx 0$ for $\alpha \neq \beta$), the teleology-weighted probabilities are

$$P_T(\alpha) := \frac{D_T(\alpha, \alpha)}{\sum_\gamma D_T(\gamma, \gamma)}. \tag{14}$$

Equations (13)–(14) generalize the usual consistent/decoherent histories framework by introducing a final weighting operator ρ_f.

4.2 Ethically biased Born rule as a special case

Consider a single projective measurement with outcomes $\{\Pi_i\}$ on some hypersurface, and let the pre-measurement pure state be

$$|\psi\rangle = \sum_i c_i\, |i\rangle\,, \qquad \rho_i = |\psi\rangle\, \langle\psi|\,. \tag{15}$$

Let the class operators be $C_i = \Pi_i$. Then (14) yields

$$P_T(i) = \frac{\mathrm{Tr}\left(\rho_f\, \Pi_i\, \rho_i\, \Pi_i\right)}{\sum_j \mathrm{Tr}\left(\rho_f\, \Pi_j\, \rho_i\, \Pi_j\right)} = \frac{|c_i|^2\, \langle i|\, \rho_f\, |i\rangle}{\sum_j |c_j|^2\, \langle j|\, \rho_f\, |j\rangle}. \tag{16}$$

If ρ_f is (approximately) diagonal in the measurement basis with

$$\langle i|\, \rho_f\, |i\rangle \propto \exp\Big(-\mathcal{E}_i/C\Big), \tag{17}$$

then (16) reduces to the biased Born rule (1).

A natural identification in the present theory is that $\mathcal{E}_i$ is the branch-evaluated ethical functional induced by $E(x)$ and $\Phi_c(x)$, e.g.,

$$\mathcal{E}_i \equiv -\lambda\,\Gamma_i \qquad \text{with} \qquad \Gamma_i := \langle i|\,\hat{\Gamma}\,|i\rangle\,. \tag{18}$$

5 Lorentz Covariance and Foliation Independence

The framework preserves local Lorentz covariance at the level of bulk dynamics because:

- The action (2) is a scalar functional of fields and the metric.
- Quantum evolution is governed by Tomonaga–Schwinger dynamics (9), with foliation independence ensured (sufficiently) by microcausality (10).
- The teleology operator (12) is constructed from hypersurface integrals of scalar densities (or hypersurface fluxes of covariant currents).

Teleology introduces an arrow of time through a global boundary condition (Σ_f) rather than through a local violation of Lorentz structure. In this sense the framework resembles other boundary-conditioned relativistic constructions (e.g., time-symmetric or two-boundary formulations), while retaining standard local dynamics.

6 Operational No-Signaling: Sufficient Conditions

Teleology-weighted probabilities generally depend on both initial and final operators. To prevent controllable superluminal signaling, we require that choices made in a spacelike-separated region B cannot alter marginal outcome statistics in region A in any operationally exploitable way.

We state a sufficient condition in an operator-algebraic form.

Proposition 1 (Sufficient condition for no-signaling)**.** *Consider two spacelike-separated regions A and B with commuting von Neumann algebras of observables $\mathcal{A}$ and $\mathcal{B}$ (microcausality). Suppose an intermediate measurement in B is represented by a complete POVM $\{M_b\} \subset \mathcal{B}$ with $\sum_b M_b^\dagger M_b = \mathbb{I}$ and an intermediate measurement in A is represented by $\{N_a\} \subset \mathcal{A}$ with $\sum_a N_a^\dagger N_a = \mathbb{I}$. Define class operators $C_{a,b} := N_a M_b$.*

If the final weighting operator factorizes as

$$\rho_f = \rho_f^{\bar{B}} \otimes \mathbb{I}_B, \tag{19}$$

(or more generally, if ρ_f lies in the commutant of $\mathcal{B}$ so that $[\rho_f, M_b] = 0$ for all b), then the teleology-weighted marginal distribution at A is independent of the choice of measurement in B.

Proof. Under microcausality, $[N_a, M_b] = 0$. The joint weight is

$$W(a,b) := \mathrm{Tr}\,(\rho_f\, N_a M_b\, \rho_i\, M_b^\dagger N_a^\dagger). \tag{20}$$

Summing over b and using completeness,

$$\sum_b W(a,b) = \mathrm{Tr}\,\Big(\rho_f\, N_a \Big(\sum_b M_b \rho_i M_b^\dagger\Big)\, N_a^\dagger\Big). \tag{21}$$

If ρ_f commutes with all M_b (or factorizes as in (19) so that the B-part is proportional to identity), then the normalization $\sum_{a,b} W(a,b)$ factorizes similarly and the ratio defining $P_T(a) = \sum_b P_T(a,b)$ is independent of the specific POVM choice in B. □

Remark 2 (Interpretation)**.** *Condition* (19) *is sufficient but not necessary. Physically, it states that the teleology operator is effectively* blind *to controllable local degrees of freedom in region B. In a cosmological implementation, ρ_f may be expected to be extremely coarse-grained (global) so that it approximates identity on laboratory subsystems, yielding practical no-signaling even when exact factorization fails.*

7 Equivalent Path-Integral Perspective (Measure Tilt)

The boundary-weighting approach can be re-expressed as a covariant modification of the path-integral measure. Formally,

$$\mathcal{Z} = \int \mathcal{D}\varphi \, \exp\left(\mathrm{i}S_{\mathrm{bulk}}[\varphi]\right) \, \exp\left(\lambda\,\Gamma[\varphi]\right), \tag{22}$$

where φ denotes all fields. The exponential factor $\exp(\lambda\Gamma)$ tilts the measure toward histories with larger Γ, i.e., higher teleology score, while leaving the local action unchanged.

Equation (22) provides a complementary interpretation: teleology is a global selection principle over histories rather than a local force.

8 Discussion and Testable Regimes

The framework makes two conceptually separable commitments:

1. **Field-theoretic extension:** introducing Φ_c and E as physical fields with covariant dynamics.
2. **Teleological selection:** introducing a small parameter λ that biases global solutions/histories via a boundary functional or final weighting operator.

In the limit $\lambda \to 0$, standard QFT probabilities are recovered. For $\lambda \neq 0$, deviations appear only in contexts where (i) branch structure is well-defined (decoherence) and (ii) ρ_f has nontrivial overlap structure with outcome projectors. The magnitude and operational detectability of deviations are model-dependent and can be constrained by no-signaling requirements and existing tests of quantum statistics.

A Optional Stochastic Unravelling (Non-Unitary Trajectories)

The present paper keeps bulk evolution unitary and implements teleology through a final weighting operator. An alternative (not adopted here) is an objective stochastic collapse dynamics on hypersurfaces. Such models can be written in covariant form using Tomonaga–Schwinger evolution plus a spacetime-indexed noise field and can reproduce teleology-weighted ensemble statistics via a change of measure (e.g., a Girsanov transform). In those approaches, individual trajectories are not unitary even though ensemble evolution can be well-defined. We omit details because the main text focuses on the unitary, boundary-selection implementation.

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